



DIGITAL ASSIGNMENT 3

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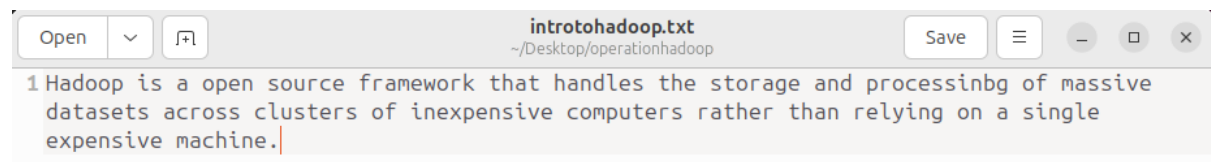
Course: Big Data Analytics Lab

Course Code: PMDS507P

Step 1: Preparing the Input Data

First, we need some data to process. A simple text file named `introtohadop.txt` was created containing a single sentence about Hadoop. This file serves as the input for our MapReduce job.

File: `introtohadop.txt` Content:



```
introtohadop.txt
~/Desktop/operationhadoop
Save
1 Hadoop is a open source framework that handles the storage and processing of massive
  datasets across clusters of inexpensive computers rather than relying on a single
  expensive machine.
```

Step 2: Creating the Mapper Script

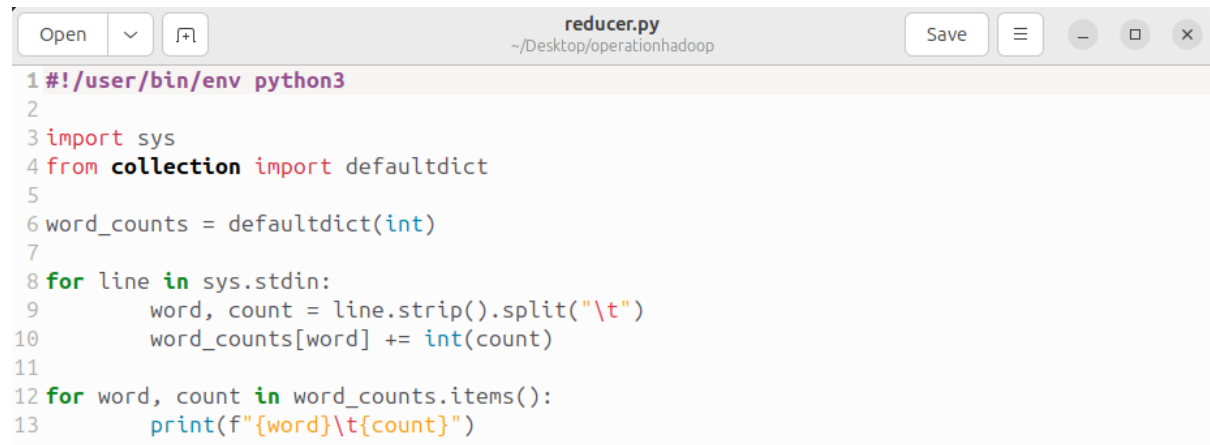
The mapper is the first phase of the MapReduce job. Its role is to read the input data line by line, process it, and emit intermediate key-value pairs.



```
mapper.py
~/Desktop/operationhadoop
Save
1 #!/user/bin/env python3
2
3 import sys
4
5 for line in sys.stdin:
6     words = line.strip().split()
7     for word in words:
8         print(f"{word}\t1")
```

Step 3: Creating the Reducer Script

The reducer receives the sorted output from the mapper. Its job is to aggregate the values for each unique key.

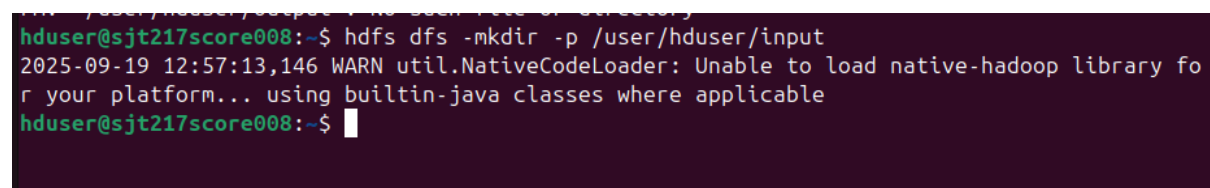
A screenshot of a code editor window titled 'reducer.py' with a file path of '~/.Desktop/operationhadoop'. The editor contains a Python script for a Hadoop reducer. The script starts with a shebang line, imports sys and defaultdict from the collection module, initializes a defaultdict for word counts, and then iterates through stdin to aggregate word counts. Finally, it prints the word and its count in a tab-separated format.

```
1#!/user/bin/env python3
2
3import sys
4from collection import defaultdict
5
6word_counts = defaultdict(int)
7
8for line in sys.stdin:
9    word, count = line.strip().split("\t")
10    word_counts[word] += int(count)
11
12for word, count in word_counts.items():
13    print(f"{word}\t{count}")
```

Step 4: Preparing the Hadoop Distributed File System (HDFS)

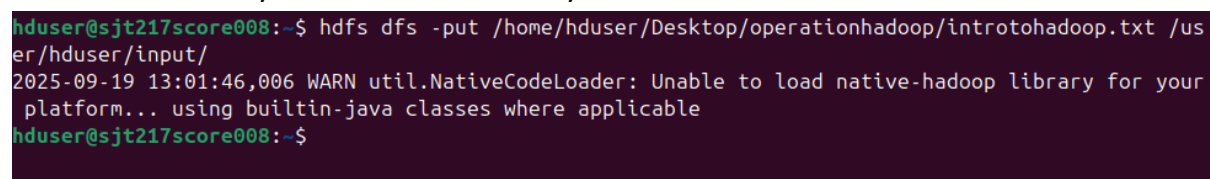
Hadoop operates on data stored within its own distributed file system, HDFS. Therefore, we must copy our local input file into HDFS.

1. Create an Input Directory: An input directory was created in HDFS to store our data file.

A terminal window showing the execution of the 'hdfs dfs -mkdir -p /user/hduser/input' command. The output shows a warning message from the NativeCodeLoader and a successful prompt for the user.

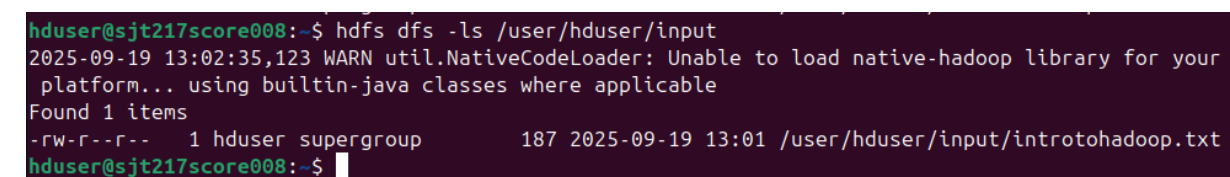
```
hduser@sjt217score008:~$ hdfs dfs -mkdir -p /user/hduser/input
2025-09-19 12:57:13,146 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hduser@sjt217score008:~$
```

2. Copy Data to HDFS: The introtohadoop.txt file was copied from the local filesystem to the newly created HDFS directory.

A terminal window showing the execution of the 'hdfs dfs -put /home/hduser/Desktop/operationhadoop/introtohadoop.txt /user/hduser/input/' command. The output shows a warning message from the NativeCodeLoader and a successful prompt for the user.

```
hduser@sjt217score008:~$ hdfs dfs -put /home/hduser/Desktop/operationhadoop/introtohadoop.txt /user/hduser/input/
2025-09-19 13:01:46,006 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hduser@sjt217score008:~$
```

3. Verify the File: We listed the contents of the HDFS directory and viewed the file's content to ensure it was uploaded correctly.

A terminal window showing the execution of the 'hdfs dfs -ls /user/hduser/input' command. The output shows a warning message from the NativeCodeLoader, a message indicating one item was found, and a detailed listing of the file 'introtohadoop.txt' with its permissions, owner, group, size, and timestamp.

```
hduser@sjt217score008:~$ hdfs dfs -ls /user/hduser/input
2025-09-19 13:02:35,123 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--  1 hduser supergroup      187 2025-09-19 13:01 /user/hduser/input/introtohadoop.txt
hduser@sjt217score008:~$
```

```

hduser@sjt217score008:~$ hdfs dfs -cat /user/hduser/input/introtohadoo.txt
2025-09-19 13:06:07,592 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Hadoop is a open source framework that handles the storage and processinbg of massive datasets ac
ross clusters of inexpensive computers rather than relying on a single expensive machine.
hduser@sjt217score008:~$

```

Step 5: Making Scripts Executable

For Hadoop Streaming to run our Python scripts, they must have execute permissions. The `chmod +x` command was used to grant these permissions.

```

hduser@sjt217score008:~$ chmod +x /home/hduser/Desktop/operationhadoop/mapper.py
hduser@sjt217score008:~$ chmod +x /home/hduser/Desktop/operationhadoop/reducer.py
hduser@sjt217score008:~$ cd Desktop/operationhadoop
hduser@sjt217score008:~/Desktop/operationhadoop$ ls -fs
total 20
4 introtohadoo.txt  4 .  4 ..  4 reducer.py  4 mapper.py
hduser@sjt217score008:~/Desktop/operationhadoop$ ls -l
total 12
-rw-rw-r-- 1 hduser hduser 187 Sep 19 12:34 introtohadoo.txt
-rwxrwxr-x 1 hduser hduser 132 Sep 19 12:47 mapper.py
-rwxrwxr-x 1 hduser hduser 269 Sep 19 12:47 reducer.py
hduser@sjt217score008:~/Desktop/operationhadoop$

```

Step 6: Running the Hadoop Streaming Job

With everything set up, the MapReduce job is launched using the `hadoop-streaming` JAR file. This command tells Hadoop where to find the input data, where to store the output, and which scripts to use for the mapper and reducer tasks.

```

hduser@sjt217score008:~$ hadoop jar /home/hduser/Hadoop/share/Hadoop/tools/lib/Hadoop-streaming-3.3.1.jar -input /user/hduser/input/introtohadoo.txt -output /user/hduser/output -mapper /home/hduser/Desktop/operationhadoop/mapper.py -reducer /home/hduser/Desktop/operationhadoop/reducer.py

```

Display the results:

```

hduser@sjt217score008:~$ hdfs dfs -cat /output/part-r-00000
2025-10-03 13:00:29,292 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Hadoop 1
a 2
across 1
and 1
clusters 1
computers 1
datasets 1
expensive 1
framework 1
handles 1
inexpensive 1
is 1
machine. 1
massive 1
of 2
on 1
open 1
processinbg 1
rather 1
relying 1
single 1
source 1
storage 1
than 1
that 1
the 1

```